

DMS DEVELOPMENT

Documentation

Abstract

In this module I learn about DMS project structure and database structure

Dhruvil Dobariya

dhruvildobariya21@gmail.com

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DATABASE STRUCTURE

1.1 TYPES OF DATABASE

- We have three types of database.
  - Company level
  - System level
  - Application level

Company level:

- It's contains two types of tables,
  - Company 's tables
  - Year's tables
- It format is: dms_companyId_financialyr
- Ex: dms_000197_2324, dms_000198_2324, dms_000197_2425,

System level:

- It contains system tables.
- It contains default data, which is shared by all clients.
- It is named as a dmssys in database.

Application level:

- It contains two type of table,
  - User detail tables
  - Company detail tables
- It contains relationships between user and company tables.
- In DMS we marge this level with System level.
- Ex:
  - SMH01, SMH02, SMH03…

1.2 TABLE NAMING CONVENTION

- We take first character based on table type,
  - If Company then, C
  - If Year then, Y
  - If System then, S
- Then we take second character based on grouping of table,

- If Master then, M
- If Transaction then, T
- If Setup then, S

- After we take third character based in sub grouping of table,
- After we take two character for sequence.

Example:

- YTT41
  - Y means Year level table
  - T means transaction group
  - T means Transaction Main sub group
  - 41 is sequence for table
- YMA01:
  - Y mean Year level table
  - M means Master group
  - A means Account Master sub group
  - 01 is sequence
- CMM01:
  - C means Company level table
  - M means Master group
  - M means General Master sub group
  - 01 is sequence
- YLL01:
  - Y means Year level table
  - L means Log group
  - L means Log sub group
  - 01 is sequence for table
- We can't take last three character as duplicate across the all databases.

1.3 Table Naming Convention

- We have three types of fields in table,
  - Normal Field
  - Common Field
  - Unique DMS Field

Normal Field:

- First, we take last three character of table name.
- Then we put F character for Filed.
- After we take two characters for sequence

Example:

- If we have table YMM41, then fields are,
  - M41F01, M41F02, M41F03…

Common Field:

- We have some common fields which is used in most of table so we should use separate naming convention for those fields,
- First three character same as normal field
- After three character we should use based on this table

Field Name	Character
DealerId	C01
SalesmanId	C02
VoucherType	C04
SubVoucherType	C05
UserId(Creation)	C06
DateTime(Creation)	C07
UserId(Modification)	C08
DateTime(Modification)	C09

DMS Unique Field:

- We have some unique field which is not in cloud Miracle but it is exist in DMS.
- First three character same as normal field
- After we take X as third character
- Then we take two characters for sequence.

Example:

- If we have YTT41 table and we want to create name for those fields which is not in Cloud Miracle but exist in DMS then,
- Name becomes T41X01

1.4 User Field and Expanse Detail Table

Table:

- We normally create new table for User field / Expanse for any existing master / transaction group of table.
- Naming of that table will be: Main Table name + “U/E”

Example:

- User field table for account master will be YMA01U whose main table is YMA01
- Expense table for item wise expense will be YTT02E whose main table is YTT02

Field:

- First field we should take last three character of table name
- After we use “U/E” respectively for User field and Expense.
- Then we take two characters for sequence.

Example:

- For User field YTT02 is T02U01.
- For User field YMA01 is A01U01.

1.5 Pseudo Name

- For security purpose we should use pseudo name in GUI.
- First we take first three character as it is.
- Then third is character is base on type of field,
  - If common field C, then 0
  - If normal field F, then 1
  - If DMS unique field X, then 2
- Then we use two character for sequence.

Example:

- T41C01 then, T41001
- T41F01 then, T41101
- T41X01 then, T41201

1.6 Datatype Guidelines

- If master should have less than 65,000 record then we use SMALLINT instead of INT in Primary Key.

- If any field would be contains two character or less than two character then we must use Char(1) and Char(2).
- Null value not allow in any field, we must try to set appropriate default value if possible.
- Don't add comment into field.

1.7 MySQL Guideline

1.7.1 Stored Procedure:

- Parameter name start with small 'p' with underscore (_)
- Ex: p_B01F01, p_B02F02
- Use Proper Datatype for Parameter
- Variable name start with small 'v' with underscore (_)
- Ex. v_UserId;
- Use Default Value if possible
- Ex. v_UserId INT DEFAULT 0;
- Don't Used and avoid @Session Variable use local variable.
- Drop temporary table after usage are complete.

1.7.2 Select Statement:

- Don't Used and avoid COUNT (*) used COUNT with Key Column.

Format:

```
SELECT
        Fields
FROM
        Table Name (Alias)
        JOIN
WHERE
        Condition
Group BY
        Fields
Order By
        Fields
```

Where Clause

- Used Primary key or Index Column First
- Date Rang using between must in bracket
- Apply filter with proper sequence , like,
    - Dealer Id
    - Salesman Id

- Date
- Sub voucher type

Join:

- Used Column they belong Primary Key or Index
- Used Column with same data type
- If we should do multiple join on same table then, we must give appropriate alias.
- Ex:
  - Left Join YMP16 P16A on P16A. P16F01 = P01F020
  - Left Join YMP16 P16B on P16B. P16F01 = P01F024

Order By Clause

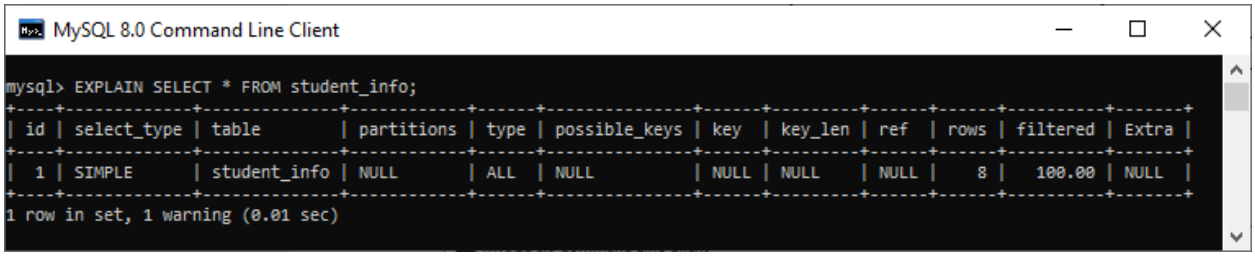
- Used only necessary
- Used only required column

Unique Row

- Must use Limit 1 if we want to get single record in API.

Explain with SELECT

- The EXPLAIN keyword is used to obtain information about how our SQL databases execute the queries in MySQL.



- id: The sequence number of the step in the query execution plan.
- select_type: The type of SELECT query (SIMPLE, PRIMARY, SUBQUERY, etc.).
- table: The name of the table being referenced in the query step.
- partitions: The partitions involved in the query step (if partitioned tables are used).
- type: The join type or access method used (e.g., "index", "all", "range", etc.).
- possible_keys: The indexes that could potentially be used for the query.
- key: The index actually chosen for the query step.
- key_len: The length of the index used.
- ref: The columns or constant values used with the index.
- rows: The estimated number of rows the query will examine or return.

- filtered: The percentage of rows filtered by the table condition.
- Extra: Additional information or notes about the query execution plan.

1.7.3 Temporary Table:

- Used when required.
- Always take required columns.
- Use primary key or index for batter performance.
- Don't restructured in between operation, it's impact on performance.
- Drop after usage.

1.7.4 Other guideline

Data Duplication

- Column Verification and avoid same data column

Column Alias

- Used Column Alias only necessary

Function with WHERE

- Used only necessary

Find_In_Set()

- Don't Used and avoid find_in_set() functions.

Commenting:

- We should add comment in stored procedure and function.
- Like parameter name, explanation of some group of statements…

1.8 Important Table

Table Name	Original Name
SMH01	Company Details
SMH02	User Details
SMH03	Year Details
SMZ01	Dropdown Master
SMZ02	Company Master
SMZ03	State Master

DMS Development

SMZ09	Company / Year Database structure changes details
SMZ11	Database Table Definition Detail
SMZ12	Database Table Field Definition Detail
SMZ13	Parent Child Master
SMZ14	Account Group Master
SMZ24	Activity Master
SMZ25	Manu Master
SMZ26	User Group Master
SMZ27	User Master
SMZ28	User Login Details
SMX01	Setup Configuration
SMX02	Setup Main Category Detail
SRR01	Template Field Master
CMB09	Activity Master
CMB10	Manu Master
CMU01	User Group Master
CMU02	User Master
YMA03	Salesman Master
YMA04	Route Master
YMA05	Dealer – Salesman Mapping
YMA06	Party Master
YMA07	Country Master
YMA08	State Master
YMA09	District Master
YMA10	Taluka Master
YMA11	City Master
YMA12	Zone Master
YMA13	Region Master
YMA14	Area Master
YLL01	User Master Log